

Alperen Demirci

Senior Artificial Intelligence Engineering Student

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EDUCATION

B.Sc. Artificial Intelligence Engineering · Hacettepe University, Ankara 2022 – 2026 (Expected)

- **GPA:** 3.58 / 4.00 **Rank:** 3rd in cohort
- **Relevant Coursework:** Computational Photography, Computer Vision, Information Retrieval, DSA, Statistics & Probability, Machine Learning, Ethics for AI, Principles of AI, Theory of Computation, Automata Theory

EXPERIENCE

Candidate Engineer — Computer Vision · ASELSAN Feb 2025 – June 2026

- Built bidirectional image translation with a custom GAN; ranked 3rd/40 in internal engineering contest. (AYetenek3)
- Deployed object detection tracking pipeline on real time pipeline. (AYetenek4)

Computer Vision Intern · ASELSAN Jul – Aug 2025

- Achieved same object detection performance at 5× **lower compute**; cut inference latency 2× via TensorRT & ONNX.

Undergraduate Researcher — Multimodal LLMs · Hacettepe / Koç University Feb 2025 – Feb 2026

- Supervised by Prof. Erkut Erdem & Prof. Aykut Erdem (DREAM program).
- Proposed a training-free hallucination benchmark for Medical MLLMs; evaluated 12+ architectures (Qwen-Thinking, InternVL3.5, MedGemma).
- Paper accepted at **CVPR 2026 MedReasoner Workshop**, poster presentation at 3-4 June 2026, Denver, Colorado. [🔗](#)

Data Science Intern · Jotform Aug – Sep 2024

- Built a text-prompt avatar personalizer tool — ranked 1st/60 among all interns; shipped to production in Jotform Agent Personalization.

Research Intern — Financial Vision · Hacettepe University Computer Vision Lab Jun – Aug 2024

- Trained ResNet18 using self-supervised learning via Lightly for chart-pattern classification; achieved **F1 = 0.93** on 1,250 images. Constructed CPVD dataset, uploaded to HuggingFace for common use at [\[here\]](#). (Paper under revision)

Teaching Assistant — Data Science Lab · Hacettepe University Oct 2024 – Jan 2025

- Prepared lab materials and delivered recitations for AIN214 under Prof. Nazli Ikizler Cinbis.

SELECTED PROJECTS

- **EOMMData - Earth Observation Multimodal Multiview VQA Data Generator** — e2e data collection pipeline for collecting multiview remote sensing visual question answering dataset. For each given location, generates questions based on OSM data and retrieves the satellite view & street view images from Sentinel-2 satellite and IGN, ESRI, NAIP remote sensing image sources. [\[code\]](#)
- **NOTA-Bench - Benchmark Repository for Multimodal LLM's on VQA** — Benchmark pipeline for multimodal medical models using inference engines as vLLM, OpenRouter. [\[code\]](#) (CVPR 2026 MedReasoner Workshop)
- **SSL for Financial Pattern Recognition** — SimCLR / BYOL / Barlow Twins on ResNet18; pre-trained on 10 yrs NIFTY50 (5-min). Detects H&S, flags & more. (Paper under revision)
- **RGB → Thermal Translation + Super-Resolution** — Caltech Aerial RGBT Dataset (ECCV 2024); Achieved **17.0 PSNR / 0.53 SSIM** on RGB to Thermal image translation. [\[code\]](#)
- **SingleLDR2HDR** — MobileNetV3 encoder-decoder for tonemapped HDR prediction using single LDR image; **15.2 PSNR / 0.87 SSIM**. [\[code\]](#)
- **Earthquake Motif Mining** — SAX/PAA + GPU-STUMPY on STEAD dataset for seismic pattern discovery across 3-channel signals. [\[code\]](#)

TECHNICAL SKILLS

ML / DL PyTorch, HuggingFace Transformers, OpenCV, scikit-learn, Lightly

Deployment TensorRT, ONNX, Docker, vLLM

Research Multimodal LLMs, GANs, Object Detection/Tracking, Self-Supervised Learning, Image-to-Image Translation

AWARDS & ACHIEVEMENTS

- 1st/60 — Jotform Internship Award, Best Data Project 2024
- 3rd/40 — AYetenek3 ASELSAN Project Contest
- 2nd/29 — HU AI Club Finance Datathon
- 10th/100 — OPC (competitive programming contest) 2024 by inzva
- 10th/150 — ALPOC (competitive programming contest) 2025
- 19th/92 — ITU AI Club Datathon (geo-localization, avg. Haversine dist. 0.20)
- Final Nominee — Midas Talent League 2026